



Sherif's Task — UML Class Diagram Fix & Update

Project: Hotel Booking System

Tool: draw.io (or any UML editor)

Assigned To: Sherif

Role: UML Diagram Designer



Your Objective

The current UML Class Diagram for our Hotel Booking System has **several missing classes, incomplete methods, and missing relationships** compared to what is actually implemented in the code.

Your job is to **open the existing diagram, fix all issues listed below, and submit the corrected version.**

[!IMPORTANT] The existing diagram file is: `F:\Java Project\Copy of java project.drawio.pdf`
Re-draw or update it based **exactly** on the fixes described below.



Issue #1 — Missing **Receptionist** Class (CRITICAL)

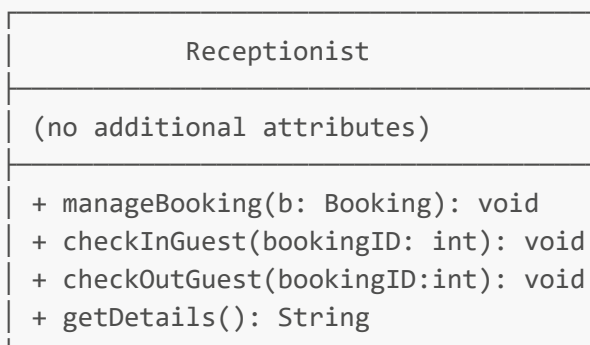
Problem: The diagram currently shows **Admin** and **Customer** inheriting from **User**, but the **Receptionist** class is completely missing — even though it exists in the actual code.



Fix: Add **Receptionist** Class

Add a new class box for **Receptionist** that **inherits from User** (use a generalization/inheritance arrow pointing up to **User**).

The class must contain exactly:



Relationship:

User  Receptionist (Inheritance / Generalization)

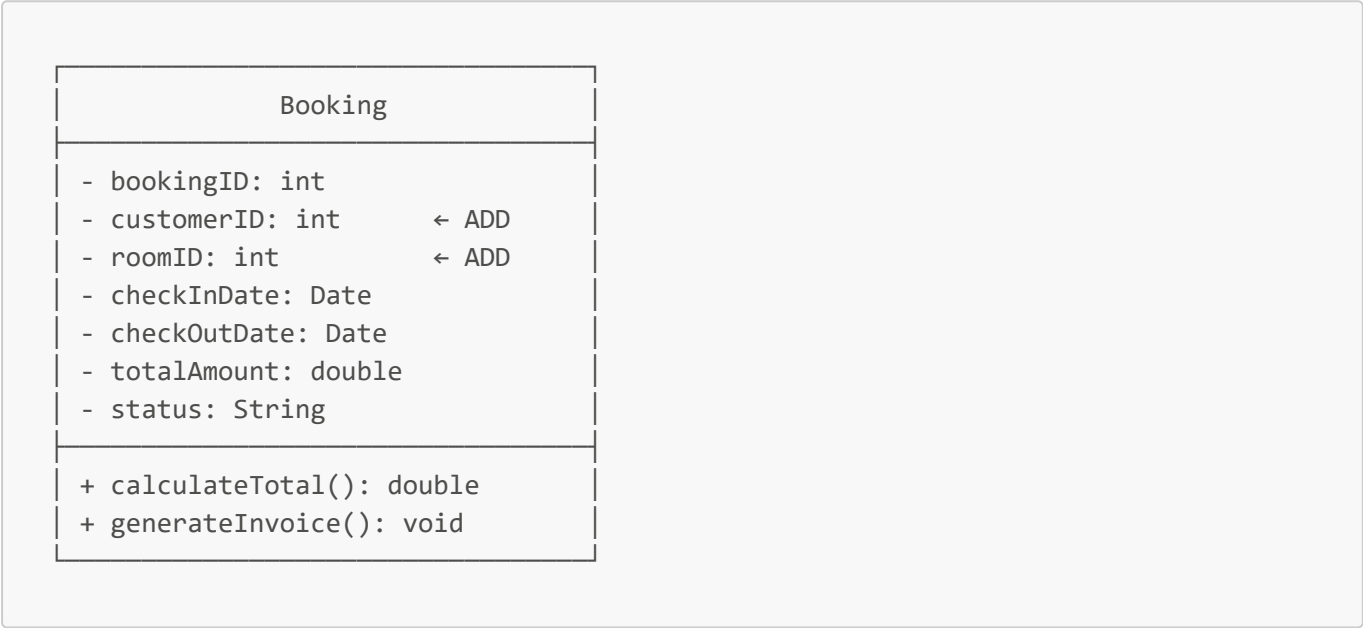
Placement: Place **Receptionist** next to **Admin** and **Customer** under the **User** class.

✖ Issue #2 — **Booking** Class is Missing Relationships

Problem: The **Booking** class does not show a link to **Customer** (who made the booking) or to **Room** (which was booked).

✔ Fix: Update **Booking** Class Attributes

Add the missing foreign-key attributes inside the **Booking** class box:



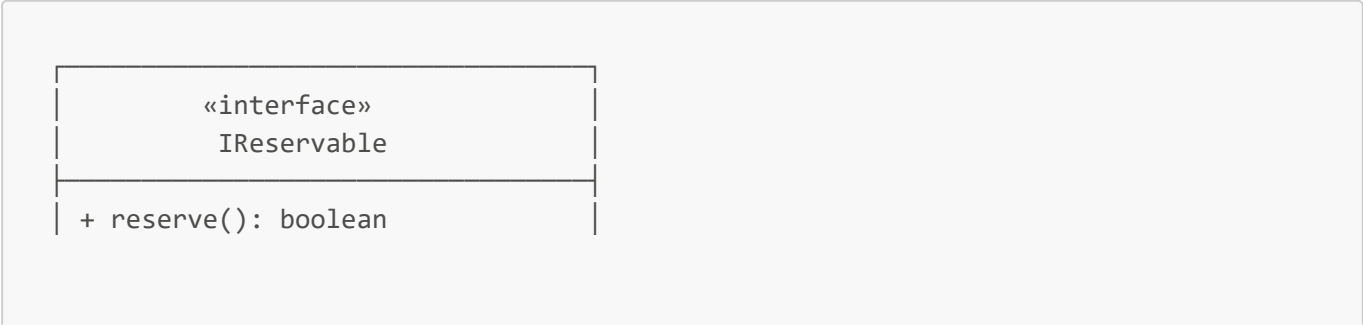
✔ Fix: Draw Two Association Arrows from **Booking**

Arrow	From	To	Label	Multiplicity
Association	Customer	Booking	"makes"	1 → 0..*
Association	Booking	Room	"reserves"	0..* → 1

✖ Issue #3 — **IReservable** Interface is Incomplete

Problem: The **IReservable** interface has two blank method entries and does not show which class implements it.

✔ Fix: Complete the Interface Box



```
+ cancelReservation(): boolean
```

Remove the two empty lines. The interface should only have those 2 methods.

✅ Fix: Show Which Class Implements It

Draw a **dashed arrow with an open triangle** (- - - ▷) from **Booking** to **IReservable**:

```
Booking - - - ▷ IReservable (Realization / Implementation)
```

❌ Issue #4 — **BookingDAO** is Missing Methods

Problem: The current **BookingDAO** only shows 2 methods, but the real implementation has more.

✅ Fix: Update **BookingDAO** Class Box

```

Booking (DAO)
+ createBooking(b: Booking): boolean
+ getBookingsForCustomer(customerID: int): List
+ getAllBookingsDetailed(): List          ← ADD
+ cancelBooking(bookingID: int, roomID: int): boolean ← ADD
+ deleteBooking(bookingID: int, roomID: int): boolean ← ADD
+ calculateRevenue(): double              ← ADD

```

❌ Issue #5 — **RoomDAO** is Missing Methods

Problem: **RoomDAO** only shows 2 methods, the real class has more.

✅ Fix: Update **RoomDAO** Class Box

```

Room (DAO)
+ getAllRooms(): List<Room>
+ getAvailableRooms(): List<Room>          ← ADD
+ addRoom(r: Room): boolean                ← ADD
+ deleteRoom(roomNumber: String): boolean ← ADD
+ updateRoomStatus(roomID: int, status: boolean): boolean

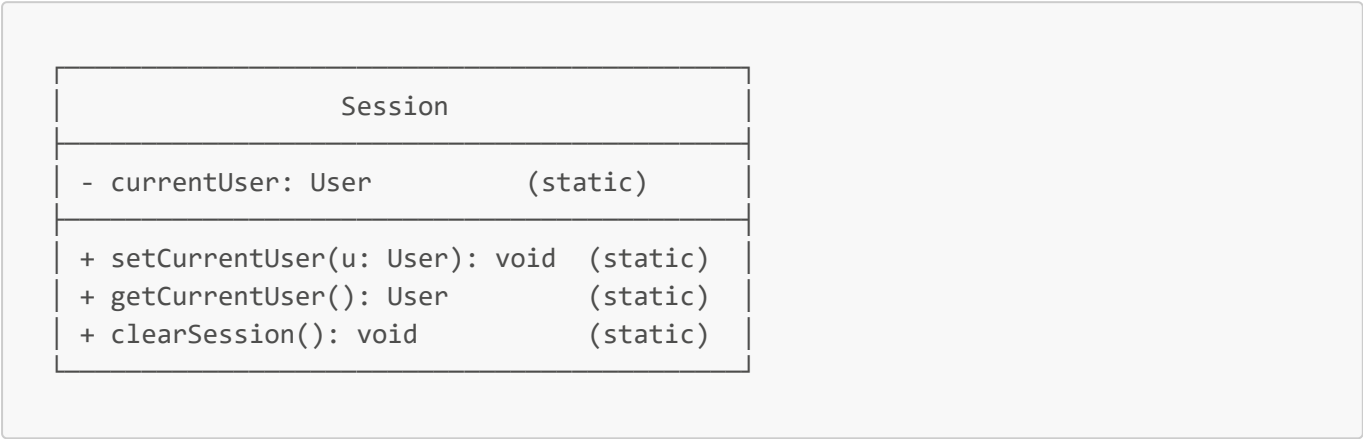
```

✖

Issue #6 — Missing Session Utility Class

Problem: The system uses a Session class to track the currently logged-in user throughout the application. It is missing from the diagram entirely.

✔ Fix: Add Session Class in the "Database & Utilities" Section



✖

Issue #7 — Multiplicity Labels Are Unclear

Problem: The numbers 1, 0..*, 1..* appear floating in the diagram without being clearly attached to specific relationship arrows.

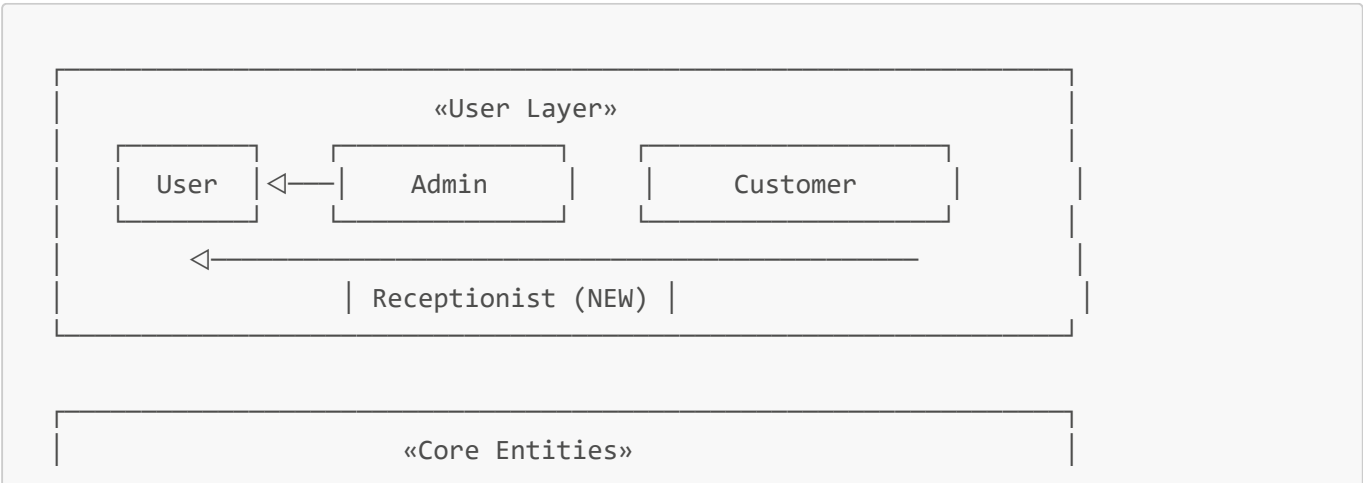
✔ Fix: Attach Correct Multiplicity to Each Arrow

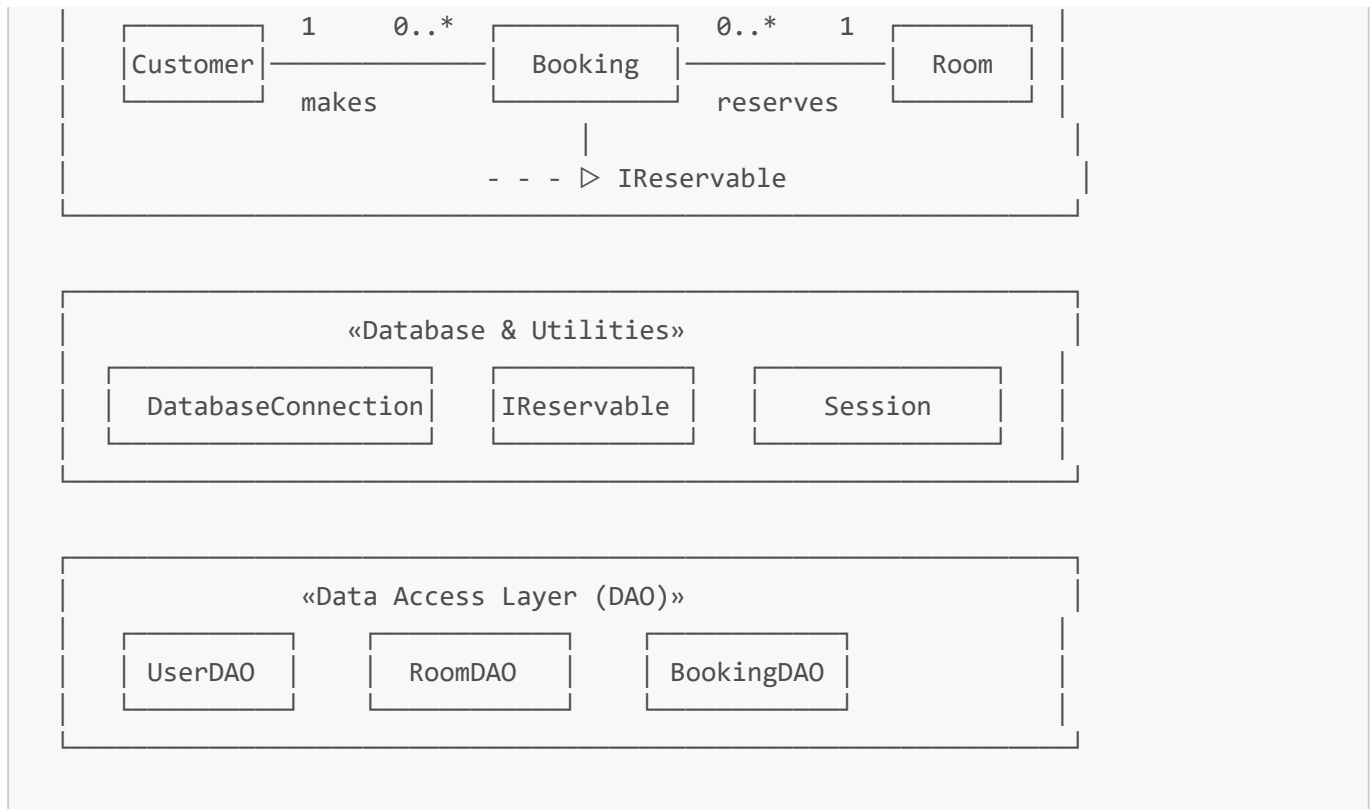
Relationship	From Side	To Side
User → Booking	1	0..*
Room → Booking	1	0..*
Admin → Room (manages)	1	0..*

🔍

Final Diagram Structure Overview

After all fixes, your diagram must have this structure:





✓ Completion Checklist

Before submitting the updated diagram, check every item:

- ☐ Receptionist class added with correct inheritance from User
- ☐ Booking class has customerID and roomID attributes
- ☐ Arrow drawn: Customer → Booking (1 to 0..*)
- ☐ Arrow drawn: Booking → Room (0..* to 1)
- ☐ IReservable interface has exactly 2 methods (no empty lines)
- ☐ Booking → IReservable realization arrow drawn (dashed)
- ☐ BookingDAO updated with all 6 methods
- ☐ RoomDAO updated with all 5 methods
- ☐ Session class added in the Utilities section
- ☐ All multiplicity labels are clearly attached to their arrows
- ☐ Diagram is clean, aligned, and readable

[!NOTE] You can use **draw.io** (free online at <https://app.diagrams.net/>) to edit the diagram. Export the final version as both **.drawio** (editable) and **.pdf** (for submission).

Good luck, Sherif! 🚀